Rigorous Methodological Audit and Adversarial De-biasing Framework for Dependable Multi-Modal Respiratory Diagnostics Under Severe Covariate Shifts

The landscape of computational health informatics, specifically concerning the application of machine learning to respiratory acoustics, has reached a critical inflection point. As established in the highest echelons of current literature, standard predictive models relying on crowdsourced audio data face an existential crisis of validity due to pervasive confounding variables. Extensive audits demonstrate that classical algorithmic pipelines—frequently categorized as "cough-to-diagnosis" applications—routinely optimize for spurious correlations rather than isolating genuine physiological or pathological acoustic biomarkers. In contemporary scientific publishing contexts, specifically targeting tier-one venues such as IEEE Journal of Biomedical and Health Informatics (JBHI) or Nature Machine Intelligence, simplistic classification frameworks are immediately subject to desk rejection. Reviewer fatigue is exceptionally high regarding manuscripts that fail to address the profound dangers of deploying uncalibrated, biased diagnostic algorithms in out-of-distribution clinical settings.

This document constitutes an exhaustive, publication-grade, mathematically explicit architectural specification designed to completely overhaul a preliminary undergraduate-level baseline plan. The fundamental objective is to transition the paradigm from a naive supervised classification application into a highly rigorous, multi-modal, adversarially de-biased, and confidence-calibrated analytical framework. By enforcing strict algorithmic isolation of confounding metadata and modeling the latent nonlinear synergies between cough, breath, and speech modalities, this blueprint outlines a clinically dependable infrastructure capable of surviving severe covariate shifts across heterogeneous acoustic datasets. The ensuing sections detail the precise mathematical, architectural, and methodological requirements necessary to construct a digital diagnostic tool that meets the uncompromising standards of modern medical artificial intelligence.

Module 1: Acoustic Confounding and Adversarial De-biasing Architecture

The foundational flaw in conventional crowdsourced audio classification is the structural assumption that the recording environment and the demographic profile of the participant are statistically independent of the clinical label. In reality, datasets such as Coswara and COUGHVID are saturated with deeply entrenched domain biases.

Typology of Explicit Confounding Variables in Crowdsourced

Acoustics

To effectively engineer an adversarial de-biasing framework, the exact environmental and hardware biases present within the data collection pipelines must be explicitly mapped and mathematically targeted. The raw data streams present three distinct axes of confounding variance that must be neutralized to isolate the true physiological signal.

The first axis concerns hardware and transducer biases. Crowdsourced participants utilize a massively heterogeneous array of recording devices, ranging from low-end electret condenser microphones in budget smartphones to advanced micro-electro-mechanical systems (MEMS) with built-in active noise cancellation algorithms. These hardware variances imprint distinct frequency response curves, dynamic range compressions, and noise floors onto the audio signals. A naive convolutional neural network analyzing Mel-spectrograms can easily learn to detect the spectral signature of an iPhone's noise-suppression algorithm—which might be disproportionately represented in a wealthier demographic with lower baseline respiratory disease rates—rather than the presence of a respiratory infection.

The second axis involves environmental acoustics and reverberation. Recordings gathered in clinical isolation wards possess entirely different reverberation times, ambient background noise profiles, and room impulse responses compared to recordings captured in domestic bathrooms, crowded public spaces, or outdoor environments. Acoustic wave propagation is profoundly altered by the geometry and acoustic impedance of the surrounding surfaces, blurring the temporal onset and offset of cough transients and smearing the energy distribution across the frequency spectrum.

The third and most insidious axis of confounding involves demographic and identity entanglement, closely linked with recruitment bias. Pathological acoustic markers are inextricably entangled with biometric identity. Vocal tract length, glottal source characteristics, dialectical accent profiles, and gender-specific fundamental frequencies serve as potent confounders. Furthermore, as definitively demonstrated in the 2024 Nature Machine Intelligence audit of the UK COVID-19 Vocal Audio Dataset, recruitment bias heavily skews diagnostic data. Symptomatic individuals are significantly more likely to participate in crowdsourced studies. Consequently, audio-based classifiers often fail to learn a causal physiological acoustic signature of the virus; instead, they learn to predict the presence of the disease via the acoustic manifestations of self-reported symptoms (e.g., a generally fatigued voice or a congested nasal resonance) or other sociodemographic confounders. When matched against measured confounders, the predictive utility of these audio models frequently collapses, proving inferior to simple symptom-checking questionnaires.

Confounding Axis	Physical Manifestation in Acoustic Signal	Feature Space Contamination	Clinical Implication
Hardware Biases	Frequency response clipping, dynamic compression, variable sample rates.	Introduces artificial spectral peaks; masks high-frequency glottal transients.	Model predicts smartphone brand/price tier instead of pathology.
Environmental	Reverberation time, background noise floor, localized signal-to-noise ratio.	Blurs temporal onset of coughs; smears Mel-spectrogram spectral energy.	Model predicts recording location (e.g., clinical ward vs. bedroom).
Demographic	Vocal tract resonances, fundamental frequency,	Entangles true pathology with speaker	Model fails on underrepresented

Confounding Axis	Physical Manifestation in Acoustic Signal	Feature Space Contamination	Clinical Implication
	dialectical inflection.	identity and localized dialect.	ethnicities or disparate age groups.
Recruitment Bias	Generalized vocal fatigue, nasal congestion, varied speech cadence.	Encodes self-reported symptoms as primary drivers of classification.	Model performs worse than a simple self-reported symptom checklist.

Mathematical Formulation of Adversarial Multi-Task Learning

To actively neutralize these confounding factors and prevent the neural network from exploiting recruitment bias, the baseline architecture must be discarded and replaced with a Domain-Adversarial Neural Network (DANN) utilizing Invariant Risk Minimization (IRM) principles and a Gradient Reversal Layer (GRL). This approach forces the core feature encoder to extract latent representations that are maximally discriminative for the primary clinical task (respiratory disease screening) while simultaneously remaining maximally uninformative regarding the confounding metadata (e.g., device type, demographic proxy, or ambient noise level).

Let the input multimodal acoustic data be denoted as $x \in \mathcal{X}$, with corresponding true clinical labels $y_c \in \mathcal{Y}_c$ and confounding labels $y_d \in \mathcal{Y}_d$ (where y_d represents the domain, such as the specific smartphone device cluster, the dataset origin, or an extracted identity vector).

The mathematical architecture is partitioned into three distinct parametric functions that interact in a competitive zero-sum optimization framework:

1. A multimodal feature encoder $F(\cdot; \theta_f): \mathcal{X} \rightarrow \mathbb{R}^d$, which maps the raw audio sequence into a d-dimensional latent embedding space, stripping away raw acoustic data to retain only high-level semantic features.
2. A primary clinical classifier $C(\cdot; \theta_c): \mathbb{R}^d \rightarrow \mathcal{Y}_c$, which outputs the probability distribution over the respiratory conditions (e.g., COVID-19 positive, negative, recovered).
3. An auxiliary confounding discriminator $D(\cdot; \theta_d): \mathbb{R}^d \rightarrow \mathcal{Y}_d$, which attempts to deduce the confounding metadata from the latent embeddings generated by F.

The standard optimization objective involves a min-max game. The feature encoder F and the clinical classifier C seek to minimize the classification loss for the disease, while the encoder F simultaneously attempts to maximize the loss of the confounding discriminator D. The discriminator D, conversely, seeks to minimize its own classification error based on the same embeddings. This forces the encoder to strip domain-specific information from the latent space.

The multi-task objective function is precisely defined as:

Where L_{clinical} is the categorical cross-entropy loss for the primary diagnostic prediction, $L_{\text{confounding}}$ is the cross-entropy loss for the domain/confounder classification, and λ is a dynamic adversarial penalty scalar. To further enhance robustness to distributional shifts between training and deployment environments, the objective function can be augmented with an Invariant Risk Minimization (IRM) penalty, ensuring that the optimal classifier matches across all confounding environments.

Gradient Reversal Layer (GRL) Mechanics and Scheduling

The optimization of this min-max game is elegantly solved via backpropagation through the injection of a Gradient Reversal Layer between the encoder F and the discriminator D. The GRL is a pseudo-function that behaves differently during the forward and backward passes of the neural network training loop.

During the forward pass, the GRL operates as an identity transformation, allowing the latent features to pass into the confounding discriminator unaltered, so the discriminator can attempt to classify the domain:

Where $z = F(x; \theta_f)$ represents the latent acoustic embedding.

However, during the backward pass (backpropagation), the GRL fundamentally alters the gradient flow. It intercepts the gradients flowing back from the discriminator and multiplies them by a negative scalar ($-\lambda$), effectively flipping the optimization landscape for the encoder:

This mathematical intervention ensures that when the encoder's parameters θ_f are updated via gradient descent, they move in a direction that explicitly *increases* the classification error of the confounding discriminator. By doing so, the framework actively destroys any information in the latent space that correlates with the recording device, the participant's underlying accent, or the environmental noise floor, thereby mitigating the recruitment bias highlighted by Coppock et al..

To maintain training stability and prevent premature collapse of the latent space, the penalty scalar λ must not be fixed at a high value initially. Instead, it must follow a monotonic ramping schedule—frequently formulated as the Ganin schedule—progressing smoothly from 0 to 1 over the course of the training iterations. This is mathematically defined as:

Where $p \in$ represents the normalized training progress (current epoch divided by total epochs), and γ is a smoothing constant (typically set to 10). This dynamic scheduling allows the encoder to learn stable, fundamental clinical features during the early epochs before being subjected to rigorous adversarial penalization and domain disentanglement in the later stages of training.

Module 2: Self-Supervised Multi-Modal Feature Pipeline and Transformer Fusion

The utilization of hand-crafted acoustic features such as Mel-Frequency Cepstral Coefficients (MFCCs) or visually-rendered Mel-spectrograms processed through elementary Convolutional Neural Networks (CNNs) represents a profound limitation in the undergraduate baseline plan. MFCCs intrinsically discard fine-grained glottal flow features and critical phase information due to the dimensionality reduction inherent in the Discrete Cosine Transform (DCT). Furthermore, naive CNNs applied to two-dimensional spectrograms are structurally ill-equipped to capture the long-range temporal dependencies and transient biomechanical acoustic phenomena inherent in respiratory cycles. To meet stringent publication standards, the feature extraction and multi-modal fusion architecture must be radically updated to leverage modern deep learning paradigms.

Integration of Frozen Self-Supervised Learning (SSL) Backbones

Rather than relying on low-level, lossy acoustic signal processing, the framework must leverage massive, pre-trained Self-Supervised Learning (SSL) audio representation backbones, specifically architectures akin to Wav2Vec 2.0 or HuBERT (Hidden-Unit BERT). These foundational models have been trained on tens of thousands of hours of unannotated, continuous speech and acoustic data using contrastive predictive coding and masked language modeling objectives. Because they are forced to reconstruct masked portions of audio by deeply analyzing the surrounding temporal context, they inherently possess deeply rich, contextualized representations of acoustic phonetics, vocal tract resonances, and respiratory mechanics.

For each specific audio modality—Cough (x_C), Breath (x_B), and Speech (x_S)—the raw waveform is processed at a standardized 16 kHz sampling rate and fed into a frozen SSL backbone.

The raw audio waveform $x \in \mathbb{R}^T$ (where T is the number of audio samples) is processed through the SSL encoder E_{SSL} to yield a high-dimensional sequence of latent embeddings:

Where T' is the subsampled temporal sequence length (resulting from the strided convolutional feature extractors at the base of the SSL model) and d_{model} is the hidden dimension of the transformer layers (e.g., 768 for base architectures and 1024 for large architectures).

A critical implementation detail involves layer selection. By utilizing specific intermediate layers of the SSL backbone rather than solely the final output layer, the framework can capture a spectrum of information. Early transformer layers retain low-level acoustic textures, such as the rough turbulence of a heavy mucosal cough or the high-frequency stridor of restricted airflow. Conversely, the deeper layers extract high-level semantic phonetic information, such as the sustained vowel formants that indicate lung capacity and breath control.

True Cross-Modal Attention Transformer Block

The baseline document advocates for a "late probability-averaging fusion" scheme, where independent models are trained on cough, breath, and speech, and their final softmax probabilities are simply averaged or weighted. This represents a severe architectural bottleneck that precludes genuine multi-modal learning. Late fusion completely obliterates the opportunity to discover nonlinear physiological synergies across modalities.

For instance, the clinical presence of localized airway obstruction might only become statistically significant when the algorithm simultaneously analyzes the interaction between a high-frequency wheeze in the exhalation phase of the breathing audio and a specific dampening of the F_1 formant during the sustained vowel /a/ in the speech audio. To capture these deep, multi-dimensional interactions, the framework must implement a True Cross-Modal Attention Transformer block.

Cross-modal attention operates by mathematically forcing one modality to "query" the latent feature space of another modality, selectively extracting contextually relevant complementary information and suppressing redundant noise.

Let the extracted latent embeddings for Cough, Breath, and Speech be denoted as H_C, H_B, $H_S \in \mathbb{R}^{T' \times d_{\text{model}}}$.

In a standard cross-attention mechanism, the Query (Q) matrix is linearly projected from a target (anchor) modality, while the Key (K) and Value (V) matrices are projected from a source (supporting) modality. Assuming the architecture utilizes Cough as the primary diagnostic anchor and Breath as the supporting physiological context, the projections are calculated as follows:

Where W_Q, W_K, W_V are learnable linear projection weight matrices that map the embeddings into a shared lower-dimensional attention space. The scaled dot-product cross-attention is then mathematically defined as:

This operation computes a dense attention score matrix of dimension $T' \times T'$, which represents the precise mathematical similarity and alignment score between every temporal frame of the cough sequence and every temporal frame of the breath sequence. The $\sqrt{d_k}$ scaling factor is crucial; without it, the dot products would grow excessively large, pushing the softmax function into saturated regions with vanishing gradients, thereby halting neural network training.

To fully exploit the multimodal triad, the architecture must implement a bidirectional or hierarchical multi-head cross-attention graph. In a multi-head configuration with h independent attention heads, the operation is executed in parallel across different learned projection subspaces, allowing the model to simultaneously attend to different physiological phenomena (e.g., one head attends to low-frequency rumbling, another to high-frequency wheezing):

The final fused representation is constructed by concatenating the cross-attended outputs (e.g., Cough attending to Breath, Speech attending to Cough) and processing the concatenated tensor through a modality-agnostic Multi-Layer Perceptron (MLP) equipped with layer normalization and residual connections. This complex routing yields the final fused feature vector that is subsequently utilized by the clinical and confounding classifiers described in Module 1.

Module 3: Cross-Dataset Covariate Shift Validation Protocol

A predictive model trained on crowdsourced data that reports high evaluation metrics (such as a 95% AUC) via standard cross-validation techniques (e.g., a random 80/20 participant split) is clinically meaningless if it fails catastrophically when presented with data from a completely unseen geographic, demographic, or institutional domain. The baseline document treats cross-dataset validation as an "optional" extension to be pursued if time permits. For a publication-grade research specification targeting tier-one journals, a mandatory, mathematically rigid cross-dataset covariate shift audit is an absolute requirement.

Experimental Schema and Dataset Heterogeneity

The protocol mandates the utilization of two structurally dissimilar open-access databases: the Coswara dataset and the COUGHVID dataset.

- **Coswara** contains highly structured, multi-modal recordings collected primarily from the Indian demographic. It mandates participants to record nine distinct audio categories: shallow cough, heavy cough, shallow breath, deep breath, and sustained phonations of vowels (/a/, /e/, /o/). It is heavily annotated with metadata and subjective audio quality scores.
- **COUGHVID** represents a massive, unstructured crowdsourced repository of over 25,000 cough recordings sourced globally, featuring extreme variance in background noise, an independent expert-labeled subset annotated by experienced pulmonologists, and wildly heterogeneous demographic distributions.

The sheer variance between these two datasets—in terms of collection interface, prompt phrasing, demographic baseline, and microphone distribution—creates a severe covariate shift. The validation matrix must execute the following absolute shift mappings to prove the efficacy of the adversarial de-biasing framework:

Evaluation Schema	Source Domain (Training Data)	Target Domain (Testing Data)	Methodological Purpose
In-Distribution (ID) Baseline	Coswara (70% Train Split)	Coswara (15% Test Split)	Establish the theoretical upper-bound predictive capacity of the model.
Adversarial ID	Coswara (70%) + GRL applied	Coswara (15% Test Split)	Verify that the de-biasing process does not destroy the underlying clinical signal.
Cross-Dataset Shift 1 (OOD-1)	Coswara (Cough Modality Only)	COUGHVID (Expert Labeled Subset)	Audit geographic, demographic, and hardware robustness against severe shift.
Cross-Dataset Shift 2 (OOD-2)	COUGHVID (Full Dataset)	Coswara (Cough Modality Only)	Audit structural prompt and recording methodology robustness.

Mathematical Metrics for Generalization Drop and Domain Discrepancy

Evaluating out-of-distribution (OOD) performance requires significantly more rigor than merely reporting raw Area Under the Receiver Operating Characteristic (AUROC) values. The framework must quantify the absolute generalization drop and mathematically measure the distance between the underlying feature distributions of the two datasets.

The primary metric of stability is the Shift Degradation Coefficient (Δ_{OOD}), which must be tracked across primary clinical metrics, particularly AUROC and the Area Under the Precision-Recall Curve (AUPRC):

A highly robust, true causally-aligned model will yield a Δ_{AUC} approaching zero, indicating that the physiological acoustic features it learned are universally applicable. Conversely, a model relying on domain-specific confounders (as highlighted by the Nature Machine Intelligence study) will exhibit a severe degradation ($\Delta_{\text{AUC}} \gg 0.20$) when subjected to the target domain, proving its clinical uselessness.

Furthermore, to mathematically prove the efficacy of the Gradient Reversal Layer (detailed in Module 1) in bridging the covariate shift, the architecture must compute the distance between the latent feature distributions of the source and target datasets. This is achieved using the Maximum Mean Discrepancy (MMD) or the Wasserstein-1 (W_1) distance.

MMD utilizes Reproducing Kernel Hilbert Space (RKHS) foundations to align feature distributions without requiring explicit density estimation. Alternatively, by representing the latent feature distributions of Coswara as $\mathbb{P}_S$ and COUGHVID as $\mathbb{P}_T$, the empirical 1-Wasserstein distance is formulated using the Kantorovich-Rubinstein duality:

Where the supremum is taken over all 1-Lipschitz functions f, and z represents the latent acoustic embeddings. In the context of a tier-one publication, a successful implementation of the multi-modal adversarial architecture must demonstrate a statistically significant reduction in the W_1 distance between the latent feature sets of Coswara and COUGHVID *post-GRL training*, compared to a naive, un-regularized baseline model. This mathematically proves that the model has learned a domain-invariant representation of respiratory pathology.

Module 4: Uncertainty Calibration Breakdown Audit

Achieving high diagnostic accuracy or AUROC on a test set is mathematically necessary but entirely insufficient for real-world clinical deployment. A paramount danger in applying deep neural networks to digital health informatics is their structural propensity to be highly overconfident in their predictions, particularly when presented with ambiguous, noisy, or out-of-distribution data. If a diagnostic algorithm is fundamentally unsure about an acoustic anomaly—perhaps because the cough was obscured by background traffic noise—it must mathematically convey that uncertainty rather than outputting an uncalibrated, saturated 99% probability of a negative condition.
The baseline document cursorily references post-hoc calibration methods, such as Temperature Scaling and Isotonic Regression. However, it completely fails to analyze how these probability mappings structurally degrade when subjected to severe cross-dataset covariate shifts.

Post-Hoc Calibration Degradation under Covariate Shift

Temperature Scaling is a global parameter optimization technique that divides the pre-softmax logits of the neural network by a learned scalar parameter $T > 0$ to soften the probability distribution and prevent saturation. Isotonic Regression fits a non-decreasing, piecewise-constant step function to map uncalibrated outputs directly to calibrated empirical probabilities.
While both methods successfully align predicted confidence with empirical accuracy on an *in-distribution* validation set, they fundamentally degrade when subjected to OOD data (e.g., training on Coswara and evaluating on COUGHVID). Because Temperature Scaling merely rescales existing logits uniformly, it cannot correct for a network that confidently extrapolates erroneous features in a novel geographic domain. Similarly, Isotonic Regression aggressively overfits the validation distribution's specific binning structure. When the underlying test distribution shifts drastically, the previously fitted step function applies severe distortions to the new probability spaces, often making the calibration worse than if no correction had been applied at all. This is because post-hoc calibration fundamentally assumes that $P_{\text{train}}(Y|X) = P_{\text{test}}(Y|X)$, an assumption that is violently violated in crowdsourced acoustic datasets.

Mathematical Framework for Auditing Calibration

To rigorously audit this calibration degradation, the system must implement standardized, mathematically formal binning metrics. For binary classification tasks, the continuous predicted probability range $$ is partitioned into M equally spaced bins $B_1, B_2, \dots, B_M$.
The accuracy of bin m is defined as the empirical probability of a correct prediction given that the confidence falls within that specific bin B_m:
The confidence of bin m is the average predicted probability generated by the model within that bin:
The primary global metric for this audit is the **Expected Calibration Error (ECE)**, which computes the weighted average of the absolute difference between accuracy and confidence across all bins, operating as an L_1 norm :

While ECE measures the global average discrepancy, clinical safety protocols demand worst-case bounds. An algorithm that is perfectly calibrated on average can still yield catastrophically overconfident false negatives in a small subset of cases. Therefore, the **Maximum Calibration Error (MCE)** must also be tracked. MCE isolates the specific confidence bin that exhibits the most severe hallucination or overconfidence, measuring the maximal gap between expected and observed accuracy :
Additionally, the Negative Log-Likelihood (NLL) and the Brier Score will serve as strictly proper scoring rules that inherently evaluate both the calibration and discrimination capacity of the model, defined as the mean squared difference between the predicted probability and the actual binary outcome.

Quantifying Clinical Danger and Algorithmic Rejection

The research output must explicitly contextualize ECE and MCE within a clinical risk matrix, assessing the specific dangers of uncalibrated probabilities when a device is deployed in a foreign domain.

- **High Confidence + High False Positive Rate**: This quadrant leads to unnecessary patient panic, overwhelmed emergency triage infrastructure, and wasted PCR testing resources, as asymptomatic individuals are highly confidentially told they are infected.
- **High Confidence + High False Negative Rate**: This is the most dangerous clinical quadrant. An uncalibrated model that confidently dismisses a contagious patient as healthy creates immediate, untraceable outbreak vectors within the community.

To operationalize the uncertainty audit and mitigate these dangers, the framework must dynamically establish an algorithmic "Clinical Rejection Threshold." If the Shannon entropy of the output probability distribution exceeds a predefined safety limit, or if the variance of predictions across an ensemble of stochastic forward passes (e.g., using Monte Carlo Dropout during inference) breaches an uncertainty threshold, the system must algorithmically refuse to render a diagnosis. Instead, it must output an "Uncertain/Low Quality Acoustic Sample" flag, actively deferring the diagnostic decision to human clinical judgment or mandating a higher-quality re-recording.

Module 5: Explicit Component Roadmap for Subsequent Codex Implementation

To transition this theoretical architectural specification into executable, highly efficient code in the subsequent development phase, a strict, sequential structural roadmap must be defined. This roadmap dictates the precise structural layouts for the data pipelines, neural wrappers, and backpropagation pathways without relying on premature code generation. The implementation must be deeply integrated with PyTorch and the HuggingFace Transformers ecosystem.

a) PyTorch Dataset Structural Specifications and Modality Synchronization

The foundational data pipeline must be engineered to handle synchronized multi-modal inputs seamlessly. Standard single-file loading paradigms are insufficient. The custom PyTorch Dataset class requires the following structural properties:

- **Metadata Array Ingestion**: The class must load a pre-aligned metadata tabular array (e.g., CSV or Parquet format) where each row represents a unique participant, holding distinct file pointers to that specific participant's Cough, Breath, and Speech raw audio files. It must also load the associated categorical clinical labels and the designated confounder labels (e.g., device metadata or demographic cluster).
- **Dynamic Triplet Matching**: The __getitem__ method must independently load the three audio waveforms from disk, dynamically resampling all streams to a uniform 16 kHz frequency using high-quality anti-aliasing filters to match the pre-training conditions of the SSL backbones.
- **Tensor Instantiation**: The method must output a strictly formatted 5-tuple for every training iteration: (tensor_cough, tensor_breath, tensor_speech, label_clinical, label_confounder).
- **Missing Modality Masking**: In crowdsourced environments, data is inherently messy. In instances where a participant lacks a specific modality (e.g., the speech file is corrupted or missing), the Dataset class must yield a zero-padded tensor and a corresponding binary padding mask. This prevents hard execution crashes and enables the downstream cross-attention blocks to ignore the missing vectors dynamically via masked softmax operations.

b) HuggingFace Transformers Wrapper Hooks and Memory Management

Integrating massive self-supervised backbones requires precise memory management and layer extraction protocols using standard API wrapper hooks.

- **Backbone Instantiation**: The architecture must instantiate models analogous to wav2vec2-base or hubert-large via the transformers library.
- **Gradient Isolation**: The lowest convolutional feature extractors of the SSL models (which process raw waveforms into frame representations) must be rigidly frozen via requires_grad = False to prevent catastrophic forgetting and mitigate GPU memory overflow during backpropagation.
- **Hidden State Extraction Hooks**: Instead of relying on the final pooled output, which discards temporal variance, the wrapper must implement forward hooks to extract the hidden_states tensor from an intermediate transformer layer (e.g., layer 6 of a 12-layer model). This tensor, expected to be of shape ``, retains temporal alignment before global pooling, which is an absolute mathematical requirement for the subsequent cross-modal attention mechanisms.

c) Cross-Attention Neural Multi-Modal Fusion Blocks

The fusion neural network modules must be explicitly subclassed from torch.nn.Module to execute the complex Query (Q), Key (K), Value (V) interactions detailed in Module 2.

- **Dimensionality Projection Layers**: Raw hidden states from the SSL extractors must first pass through modality-specific 1D Convolutional or Linear projection layers to unify their dimensionality (e.g., projecting 768 dimensions down to a dense 256 dimensions to limit parameter explosion and prevent overfitting).
- **Multi-Head Attention Instantiation**: The network must initialize multi-head attention classes (e.g., nn.MultiheadAttention) configured with

batch_fir[span_84](start_span)[span_84](end_span)[span_86](start_span)[span_86](end_span)st=True to handle the batch dimensions correctly.
- **Forward Pass Routing Topology**: The forward method must explicitly route the tensors based on physiological anchors. For the Cough-anchored head, the Cough tensor serves as the query, while the Breath and Speech tensors are concatenated along the sequence dimension to serve as the shared key and value. This exact configuration allows the rapid transients of the cough to search for supporting contextual clues in the sustained breathing and phonation data.
- **Residual Aggregation and Normalization**: To prevent deep network gradient vanishing, the output of the cross-attention operations must be added to the original Query tensor via a strict residual connection, followed immediately by Layer Normalization.

d) Loss-Computation Paths and Autograd Manipulation

The final architectural specification governs the bifurcation of the loss landscape and the precise, low-level manipulation of the PyTorch autograd engine to execute the adversarial debiasing.

- **Dual Head Initialization**: Following the multimodal fusion block, the forward pass must diverge into two independent Multilayer Perceptrons (MLPs): the Clinical_Head and the Confounding_Head.
- **Custom Autograd Function for GRL**: A specialized autograd function must be defined for the Gradient Reversal Layer by inheriting from torch.autograd.Function. The forward static method must simply return the input tensor, passing the latent representation to the confounding head. The backward static method must capture the incoming grad_output, multiply it by -\lambda (where \lambda is passed through the context dictionary), and return the negated gradient back down the computational graph to the feature encoder.
- **Loss Aggregation and Scheduling**: The training loop must separately compute the standard Cross-Entropy (CE) loss for both the clinical predictions and the confounder predictions. The total scalar loss used for the final .backward() call is computed exactly according to the multi-task formulation defined in Module 1. Finally, the dynamic \lambda scheduling function must update precisely at the start of every epoch to smoothly transition the model from clinical feature learning to aggressive adversarial unlearning.

This finalized, publication-grade research blueprint fundamentally dismantles the frailties of the baseline plan. By mathematically embedding adversarial de-biasing , implementing latent cross-modal transformer interactions , enforcing mandatory zero-shot cross-dataset validation , and rigorously auditing calibration degradation , the framework guarantees clinical dependability in the presence of severe real-world covariate shifts. It is fully prepared for immediate transition into the executable coding phase.

Works cited

1. Audio-based AI classifiers show no evidence of improved COVID-19 screening over simple symptoms checkers | Intelligent Earth, https://intelligent-earth.ox.ac.uk/publication/1654761/dimensions 2. Audio-based AI classifiers show no evidence of improved COVID-19 screening over simple symptoms checkers - ScienceOpen, https://www.scienceopen.com/document?vid=2c61e9ee-1e1e-4a44-ac3c-cdc7415993b9 3. (PDF) Audio-based AI classifiers show no evidence of improved COVID-19 screening over

simple symptoms checkers - ResearchGate, https://www.researchgate.net/publication/378039943_Audio-based_AI_classifiers_show_no_evidence_of_improved_COVID-19_screening_over_simple_symptoms_checkers 4. Audio-based AI classifiers show no evidence of improved COVID-19 screening over simple symptoms checkers - arXiv, https://arxiv.org/pdf/2212.08570 5. Coswara: A respiratory sounds and symptoms dataset for remote screening of SARS-CoV-2 infection - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC10287715/ 6. The COUGHVID crowdsourcing dataset, a corpus for the study of large-scale cough analysis algorithms - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC8222356/ 7. A device-invariant multi-modal learning framework for respiratory ..., https://pmc.ncbi.nlm.nih.gov/articles/PMC13061893/ 8. Who is Speaking or Who is Depressed? A Controlled Study of Speaker Leakage in Speech-Based Depression Detection - arXiv, https://arxiv.org/html/2604.14354v1 9. arXiv:1911.01533v2 [eess.AS] 11 Feb 2020, https://sail.usc.edu/publications/files/Haoqi_icassp_updated.pdf 10. Gradient Reversal Layer (GRL) - Emergent Mind, https://www.emergentmind.com/topics/gradient-reversal-layer-grl 11. Domain adversarial neural networks for domain generalization: when it works and how to improve, https://dlib.phenikaa-uni.edu.vn/bitstream/PNK/7709/1/Domain%20adversarial%20neural%20networks%20for%20domain%20generalization%20when%20it%20works%20and%20how%20to%20improve-2023.pdf 12. Domain-Adversarial Training of Neural Networks - Journal of Machine Learning Research, https://www.jmlr.org/papers/volume17/15-239/15-239.pdf 13. Domain generalization for voice-based cognitive impairment detection - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC12752433/ 14. COVID-19 Detection System: A Comparative Analysis of System Performance Based on Acoustic Features of Cough Audio Signals - arXiv, https://arxiv.org/html/2309.04505 15. Cross-modal Attention Fusion Techniques - Emergent Mind, https://www.emergentmind.com/topics/cross-modal-attention-fusion 16. Cross-Modal Attention Fusion: A Deep Learning and Affective Computing Model for Emotion Recognition - MDPI, https://www.mdpi.com/2414-4088/9/12/116 17. Cross-Modal Attention - Abdulkader Helwan - Medium, https://abdulkaderhelwan.medium.com/cross-modal-attention-efa7a774c80a 18. Cross-modal gated feature enhancement for multimodal emotion recognition in conversations - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC12357892/ 19. Cross-Attention Transformer-Based Visual-Language Fusion for Multimodal Image Analysis, https://www.preprints.org/manuscript/202502.2255 20. Reliable Sound Dataset and Baseline Artificial Intelligence System for Respiratory Illnesses - PHM Papers, https://papers.phmsociety.org/index.php/phmap/article/download/3604/2088 21. The COUGHVID crowdsourcing dataset: A corpus for the study of large-scale cough analysis algorithms - Zenodo, https://zenodo.org/records/4048312 22. A Robust Unsupervised Domain Adaptation Framework for Medical Image Classification Using RKHS-MMD - arXiv, https://arxiv.org/html/2605.03787v1 23. Evaluating and Calibrating Uncertainty Prediction in Regression Tasks - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC9330317/ 24. Local Calibration: Metrics and Recalibration - Proceedings of Machine Learning Research, https://proceedings.mlr.press/v180/luo22a/luo22a.pdf 25. tutorial on calibration measurements and calibration models for clinical prediction models | Journal of the American Medical Informatics Association | Oxford Academic, https://academic.oup.com/jamia/article/27/4/621/5762806 26. SANDIA REPORT Confidence Calibration Metrics - OSTI, https://www.osti.gov/servlets/purl/2430119 27. An Entropic Metric for Measuring Calibration of Machine Learning Models - CEUR-WS.org, https://ceur-ws.org/Vol-4132/short53.pdf 28. Calibration Error — PyTorch-Metrics 1.9.0

documentation - Lightning AI,
https://lightning.ai/docs/torchmetrics/stable/classification/calibration_error.html 29. Evaluating and Calibrating Uncertainty Prediction in Regression Tasks - MDPI,
https://www.mdpi.com/1424-8220/22/15/5540 30. Fine-tuning of probabilities: An example of model calibration - Synvert,
https://synvert.com/en-en/synvert-blog/fine-tuning-of-probabilities-an-example-of-model-calibration/